ModelIt!

Teacher's Guide

The Mushroom Network Under Your Feet

Trees Share Food and Warnings Through Underground Internet

Quick Reference

Grade Level	Grade 5
NGSS Standard	5-LS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 5-LS2-1

Develop a model to describe the movement of matter among plants, animals, decomposers, and the environment.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students model matter movement through the mycorrhizal network.
Disciplinary Core Idea	**LS2.B Cycles of Matter and Energy Transfer** — Matter cycles through ecosystems including decomposers.
Crosscutting Concept	**Systems and System Models** — Students model the forest as an interconnected system.

CAST Testing Alignment

MULTIPLE CHOICE:

What happens to nutrient transfer between trees during a severe drought?

CONSTRUCTED RESPONSE:

Explain why replanting a clear-cut forest with new trees doesn't immediately restore it to its original state.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

The Wood Wide Web is a relatively recent scientific discovery. Dr. Suzanne Simard proved in the 1990s that trees communicate and share resources through underground fungal networks.

Discovery

Dr. Suzanne Simard traced radioactive carbon from one tree to its neighbors through fungal threads. She proved trees were actively sharing resources.

The Deal

Trees and fungi have a mutualistic relationship. Trees give fungi sugar from photosynthesis. Fungi give trees water and minerals they can't access alone.

Network Structure

Mother trees - the largest, oldest trees - are hubs connecting many neighbors. When they die or are cut, dozens of other trees lose their connections.

Warning Signals

Trees can send chemical alarm signals through the network when attacked by insects. Neighboring trees ramp up their defenses in response.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — **Locate the System**
Students identify leaf litter, soil moisture, fungal network activity, and nutrient transfer as components.

E — **Establish Relationships**
Students establish that organic matter and moisture both positively affect network activity, which positively affects nutrient transfer.

V — **Visualize & Model**
Students build the forest network model showing inputs driving connectivity.

E — **Evaluate Outcomes**
Students run drought and deforestation scenarios to see network collapse.

R — **Revise & Extend**
Students add soil pH to model acid rain effects on the network.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Trees talk to each other. Through mushrooms. Underground."*

Watch For

Let that statement sink in.

Slide 2: Learning Objectives

2 min

Say: *"Today we model the forest's underground internet."*

Watch For

Define mycorrhizal.

Slide 3: Big Question

2 min

Say: *"Dr. Simard proved trees share. How does it work?"*

Watch For

Show her research context.

Slide 4: LEVER Framework

2 min

Say: *"We'll model what keeps the network healthy."*

Watch For

Preview activities.

Slide 5: Activity 1: Components

8 min

Say: *"Fungi need food and water, just like us."*

Watch For

Connect to decomposition lesson.

Slide 6: Activity 2: Connections

8 min

Say: *"All positive - when conditions are good, everyone benefits."*

Watch For

Discuss mutualism.

Slide 7: Activity 3: Simulation

10 min

Say: *"Drought doesn't just hurt trees - it isolates them."*

Watch For

Students run drought scenario.

Slide 8: Discoveries

5 min

Say: *"A forest is one connected community, not separate trees."*

Watch For

Discuss implications for conservation.

Slide 9: STEM Challenge

10 min

Say: *"Design a plan to restore a damaged forest network."*

Watch For

Research mycorrhizal restoration.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Leaf Litter / Organic Matter, Soil Moisture
- **INTERNAL:** Mycorrhizal Fungal Network Activity, Nutrient Transfer Between Trees

Relationships

- **Leaf Litter to Fungal Network Activity:** POSITIVE (+) — More organic matter provides more food for fungi to grow.
- **Soil Moisture to Fungal Network Activity:** POSITIVE (+) — Fungi need moisture to grow and transport nutrients.
- **Fungal Network Activity to Nutrient Transfer:** POSITIVE (+) — More active network means more sharing between trees.

Key Discoveries

- Trees are connected underground through fungal threads
- The network allows trees to share sugar and warnings
- Drought isolates trees by shutting down the network
- Replanting a cleared forest takes decades because the network must rebuild

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Mushrooms are plants"	Fungi are their own kingdom - neither plant nor animal. They have unique properties.
"Fungi just decompose things"	Mycorrhizal fungi are PARTNERS with living trees, not decomposers.
"Trees only compete"	Trees compete AND cooperate simultaneously. The Wood Wide Web shows extensive cooperation.